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A Multi-Scale Hybrid Segmentation and MRMR Feature Selection Approach for Road Crack Detection Using XGBoost Classifier

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Abstract: Road crack detection plays an important role in the latter, which is crucial to road safety and infrastructure maintenance. This paper proposes a novel architecture embedding deep features extracted by an Improved PSP-Net and Enhanced HED Network (HED), together with Local Binary Pattern (LBP)-based texture features. The Improved PSP-Net collects multiscale contextual information in different scenarios for correct object detection, and the Enhanced HED Network enhances edge detection to more accurately classify crack boundaries. Such use of LBP-based texture features enriches the model's ability to capture small details of surface unevenness which helps in detecting intricate and complex crack morphology. In addition, it uses a Minimum Redundancy Maximum Relevance (MRMR) feature selection technique that selects the relevant and non-redundant features which increases the efficiency. A sophisticated XGBoost classifier then assigns the detected cracks into specific classes, allowing for robust and accurate crack type prediction. Experimental evaluations demonstrate that the model outperforms (99.09%) and is more robust (F1-score 99.11%, sensitivity 99.53%) than both conventional image processing methods (Gabor filtering + morphological methods) and deep learning methods (CNNs, ConvNets, optimized YOLOv8, Inception V2, DeepCrack, CrackSeg, U-HDN), suggesting that the proposed model holds wide potential impact on actual road maintenance and inspection applications.

Keywords: Enhanced HED network, Improved PSP-net, Local binary pattern, XGBoost.

1. Introduction

Crack detection in images is a critical task in infrastructure maintenance, as timely identification of cracks can help prevent structural failures and reduce repair costs [1]. This task has garnered significant attention from researchers, leading to the development of a variety of techniques that broadly fall into two categories: traditional image processing and machine learning approaches, and deep learning-based detection networks [2].

Traditional image processing and machine learning techniques focus on extracting handcrafted features from images to identify cracks. These methods rely on algorithms such as Gabor filtering, histogram-based analysis, and morphological operations. For instance, techniques like CrackTree, developed by the authors of [3], utilize geodesic shadow removal and tensor voting for crack detection, while the authors of [4] leverage histograms of gradients (HoG) to analyze edge orientations in grayscale images. These methods excel in analyzing high-resolution images with distinct crack patterns. However, their performance often diminishes in challenging environments, such as images with complex pavement textures or varying lighting conditions.

On the other hand, deep learning-based methods have emerged as powerful tools for crack detection. These techniques use convolutional neural networks (CNNs) and other advanced architectures to automatically learn and extract features from large datasets. Methods such as encoding-decoding network [5] and Jiang et al.'s MSK-UNet [6] demonstrate superior adaptability in detecting cracks, even in challenging scenarios. Despite their robustness, deep learning models often require high-quality, close-range data to achieve accurate results and may struggle with UAV-acquired imagery, where cracks occupy a smaller proportion of the image and are accompanied by significant noise.

In addition to detection, crack classification techniques aim to categorize cracks based on their characteristics. While traditional approaches, such as support vector machines (SVMs) [7] and decision trees [8], are effective for simple datasets, deep learning-based methods like CNNs [9] have shown greater promise for handling complex and large-scale data. However, these techniques are computationally intensive and require extensive labeled datasets for training.

Despite these advancements, a key challenge remains in achieving accurate and efficient crack detection and classification in complex, noisy environments, such as UAV-acquired images. This highlights the need for innovative approaches that can balance the strengths of traditional image processing with the adaptability of deep learning.

The key challenge that this work identifies is the accurate detection and classification of road cracks in UAV-acquired images, which is characterized by high noise, various environmental conditions, and fine features of cracks. Current crack detection methods, whether based on traditional image processing techniques or deep learning-based techniques, have difficulty achieving robustness and high accuracy in such a complex and noisy environment. Existing approaches typically struggle with the high variability of pavement textures, and deep learning models are not only dependent on high-quality close-range data, but they also face the problem of performance drop when used on real-world UAV imagery. In this paper, we proposed a hybrid approach, which leverage the advantages of classical feature extraction method and deep learning models to enhance accurate detection and classification of crack features in challenging noisy and complex road surface images.

The significant contributions of this work are as follows:

- Hybrid Feature Integration: The study combines deep learning features extracted from an Improved PSP-Net and Enhanced Holistically-Nested Edge Detection (HED) Network with traditional texture-based features derived from Local Binary Patterns (LBP). This integration provides a more comprehensive and robust representation of road surface images, bridging the gap between traditional and modern approaches.
- Accurate Segmentation with Hybrid PSP-Net and HED: The proposed hybrid PSP-Net and HED network enhances segmentation accuracy by efficiently identifying and delineating regions of interest (crack regions). This improves detection performance, particularly in noisy and complex environments.
- Robust Crack Classification Using XGBoost: The study employs an XGBoost classifier to predict crack types post-segmentation. This step ensures precise categorization of cracks, facilitating detailed road surface analysis for maintenance and repair planning.

By combining the strengths of traditional and deep learning methods, this paper sets a new standard for crack detection and classification in UAV-based imagery, contributing to the advancement of automated infrastructure monitoring systems.

The remainder of the paper is organized as follows: In Section 2, a literature review of crack detection methods is presented, followed in Section 3 by a description of the proposed methodology, including the hybrid model, feature extraction and classification methods. Results and discussion are given in Section 4, whereas conclusions and future work in Section 5.

2. Literature review

The literature extensively discusses various systems and methodologies for road surface inspection. The authors of [10] introduced a multifunctional road detection vehicle equipped with an inertial navigation system, laser sensors for evenness and texture measurement, a 2D laser range finder, and two cameras. However, many studies report the use of simpler recording equipment, such as smartphones mounted on vehicles, to capture pavement surface images. Despite advancements in image acquisition techniques, analysis often relies on traditional manual image processing methods, such as those performed in the CNER laboratory using the "Argus" software. This highlights the limited application of automatic image processing in certain regions.

The authors of [11] proposed and evaluates a novel approach for automatic road crack detection using a Convolutional Neural Network (CNN). The method enables classification of road cracks into three categories: longitudinal cracks, alligator cracks, and no cracks. To demonstrate its effectiveness, the proposed CNN model's results are compared to those of a pre-trained transfer learning model, Visual Geometry Group 19 (VGG-19). Numerous studies have explored deep learning algorithms for road crack detection. For example, the authors of [12] applied ConvNets to a database of 500 images to detect cracks from image patches. The authors of [13] employed YOLO V2 using smartphone-acquired images, and the authors of [14] achieved higher accuracy with optimized YOLOv8, using large-scale image databases.

In 2019, significant progress was made in adaptive segmentation methods for crack detection. The authors of [15] introduced the DeepCrack model, which achieved an F1 score exceeding 87%. However, DeepCrack's sensitivity to noise was a noted limitation. The authors of [16] developed CrackSeg, and the authors of [17] introduced ConCrack, combining adversarial networks and connectivity maps. The authors of [18] proposed U-HDN, an encoder-decoder architecture utilizing hierarchical feature learning and dilated convolution, achieving an accuracy of 97.27% with a lightweight CNN model suitable for real-time applications.

Other advancements include ensemble CNNs without pooling layers, proposed by the authors of [19], the RAO-UNet model, which improved detection accuracy and processing speed. Researchers have also demonstrated the robustness of crack detection algorithms across varying conditions. For example, the authors of [2] found that weather conditions had minimal impact on crack detection accuracy.

Transfer learning algorithms have further enhanced detection capabilities, especially with large datasets. Pre-trained CNN architectures like AlexNet, VGG-16, ResNet, and SqueezeNet have been widely applied. Wavelet transform and Radon Neural Networks (WRNN) have also proven effective, with researchers leveraging image segmentation methods such as thresholding, clustering, and edge detection to address noise and enhance accuracy.

Preprocessing techniques play a critical role in crack detection. Oliveira and Correia introduced dynamic thresholding and modified Otsu methods to distinguish crack pixels. Clustering methods, like k-means, have been utilized for feature extraction and threshold calculation, while bi-dimensional empirical mode decomposition (BEMD) has been employed for noise removal and effective edge detection.

Recent innovations, such as geodesic shadow removal and tensor voting, have further refined crack detection by building probability maps and removing noise. These techniques, while effective, often have limitations in processing time and scalability for real-time applications. Addressing these challenges remains an active area of research, aiming to create more efficient and robust road surface inspection systems.

Drawbacks of Conventional Methods:

Traditional image processing methods: These are the labour-intensive algorithms which prove incapable of scaling. They require manual contextualisation and expert knowledge, which makes them impractical for scaling and real-time applications. It is also unable to adjust to changes in lighting and road surface conditions, limiting their real-world accuracy [12].

CNN Based Methods: CNN has performed well in the controlled environment but there could be a degrade in performance in noisy and complex environments. They rely heavily on high-quality, labeled datasets, which are difficult to acquire for road crack detection, especially detection from imagery obtained via UAV [11]. This limitation degrades the applicability of CNN-based models to real-world applications.

ConvNets & YOLO V2: These models are effective in some cases, but do not generalize to new and unseen data. They also have difficulty with minor cracks and subtle characteristics, so their detection precision decreases in noise, confused images [12]. Moreover, they also have to deal with heavier computational cost and fail to scale well in real time applications, as they need larger dataset and heavier architectures.

DeepCrack and CrackSeg: Though they perform well in certain contexts, they are sensitive to noise and require large annotated datasets, making them unsuitable for real-world scenarios. They are not robust enough for practical application due to their inability to accommodate different crack structures and environmental noise 15.

U-HDN and RAO-UNet: Despite the strong performance for real-time tasks, the models still have limited adjustability to the small irregular crack pattern and noise. In some situations, such as complex road surface conditions, the performance is degraded, and they are less reliable in different environments; as a result, 18.

The approach we proposed is a hybrid of deep learning network and traditional image processing methods, thus utilizing the advantages of both analysis methods. This method combines benefits of Improved PSP-Net and Enhanced HED Network to produce a more accurate and holistic identification for the road surface by incorporating texture features including LBP. This method filters out noise and provides more precise segmentation results, improving the quality of regions of interest, especially in fields such as UAV-acquired imagery, where complex ghost targets are present. Additionally, this paper was verified but the XGBoost charge used for crack classification, which subsequently improved the predictive ability of the model for crack type and made it suitable for real-time monitoring of infrastructure.

3. Proposed methodology

The proposed methodology for road crack detection and classification comprises several key stages, including image acquisition, preprocessing, feature extraction, feature selection, and classification, each designed to enhance the system's accuracy and efficiency.

The process begins with data acquisition, where crack detection images are collected under various environmental conditions. During preprocessing, these images undergo segmentation using a Hybrid Model combining PSP-Net and HED Network, which significantly improves image quality by accurately segmenting regions of interest and preserving crucial features.

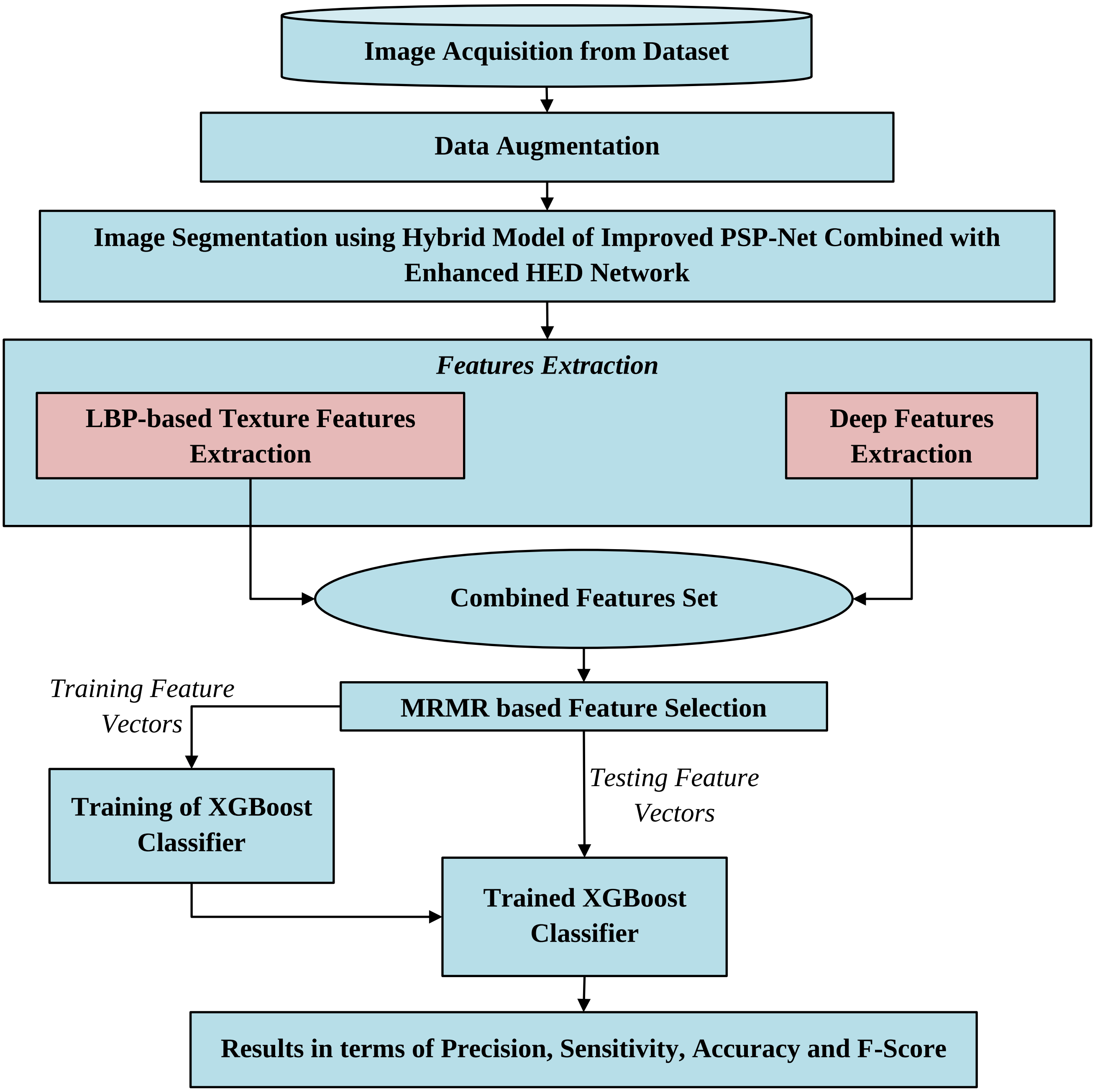

Figure. 1 Flow Diagram for Proposed Road Crack Detection

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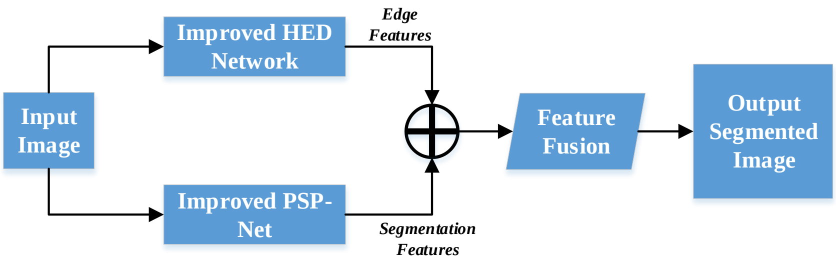


Figure. 2 Segmentation using Hybrid Model

Next, feature extraction is performed to capture relevant characteristics of the segmented images. This approach combines LBP-based texture features, which effectively represent textural patterns, with deep features derived from the Hybrid PSP-Net and HED Network, resulting in a comprehensive and computationally efficient feature set. To further refine the model, the feature selection phase utilizes the Minimum Redundancy Maximum Relevance (MRMR) technique, which selects the most relevant and non-redundant features, enhancing the classification process.

The classification phase employs an XGBoost classifier, which ensures robust crack type prediction. This systematic, multi-step framework not only improves detection accuracy but also optimizes the process for practical applications in real-world road surface monitoring scenarios.

3.1 Image acquisition from dataset

The initial step in this methodology is the acquisition of images from a publicly available dataset. This dataset contains a variety of images depicting different type of road crack,

Mathematical Formulation for Image Acquisition

Let D represent the dataset comprising N images. Each image I_i in the dataset can be denoted as:

$$D = \{I_1, I_2, \dots I_N\} \tag{1}$$

Where I_i is an image captured in RGB color space:

$$I_i = \{I_i^R, I_i^G, I_i^B\} \tag{2}$$

Where I_i^R, I_i^G, and I_i^B representing the red, green, and blue channels of the image, respectively.

Each image I_i is associated with a label y_i indicating its crack status or type of road:

$$(I_i, y_i) \in D \tag{3}$$

The labels y_i are categorical and can take values from a predefined set of classes C, where $C = \{c_1, c_2, \dots, c_k\}$ and k is the number of distinct classes (including different cracks and without crack class).

3.2 Data augmentation

Data augmentation is a crucial step in enhancing the performance of deep learning models by artificially increasing the size and diversity of the training dataset. This is particularly important in medical imaging, where datasets may be limited in size. By applying a variety of transformations to the original images, data augmentation helps in improving the generalization ability of the model, making it more robust to variations in real-world data.

3.3 Segmentation using hybrid model of improved PSP-Net combined with enhanced HED Network

The segmentation phase is critical for identifying crack-affected areas in roads. This methodology employs a Hybrid Model that combines the improved PSP-Net (IPSP-NET) with the enhanced HED Network (EHED-NET) [20]. This integration enhances segmentation accuracy and boundary delineation, addressing the complexities of road crack patterns as shown in Fig. 2.

The segmentation framework begins with the input image I_i acquired from the dataset. The image is processed through two parallel networks: the improved PSP-Net for semantic segmentation and the enhanced HED Network for edge detection.

Mathematical Formulation for Segmentation

- Feature Extraction via Improved PSP-Net: The improved PSP-Net architecture is designed to extract high-level features while maintaining spatial information. The output feature map F after passing through the improved PSP-Net is defined as:

$$F = f_{PSP}(I_i) \tag{4}$$

Where f_{PSP} represents the function of the improved PSP-Net.

- Edge Detection with Enhanced HED Network: The enhanced HED Network enhances the model's capability to detect edges in the input image. The edge map E is generated by applying a series of convolutional operations followed by an upsampling process:

$$E = g_{HED}(I_i) \tag{5}$$

Where g_{HED} indicates the operation performed by the enhanced HED Network.

- Fusing Features and Edge Information: To improve segmentation accuracy, the features extracted from the improved PSP-Net are fused with the edge information from the enhanced HED Network. This is achieved through a concatenation operation that integrates the feature map and the edge map:

$$F_{fused} = F \oplus E \tag{6}$$

This concatenated feature map contains both the high-level features necessary for segmentation and the precise edge information essential for delineating boundaries.

- Contextual Refinement: A context-aware refinement mechanism is applied to F_{fused} to enhance the segmentation output. This layer aims to refine the fused features by considering contextual information across the combined feature map:

$$F_{refined} = \sigma\left(W_{refine} * F_{fused}\right) \tag{7}$$

Where W_{refine} represents the weights of the convolutional filters used in the refinement layer, and σ denotes a non-linear activation function, such as ReLU.

- Segmentation Output: The final segmentation output S is generated using a Softmax function applied to the refined feature map, yielding class probabilities for each pixel:

$$S = Softmax\left(W_{final} * F_{refined}\right) \tag{8}$$

Here, $S \in \mathbb{R}^{H \times W \times K}$, where K denotes the number of distinct classes.

The integration of the improved PSP-Net and enhanced HED Network provides several distinctive advantages:

- Multi-Scale Feature Extraction: The improved PSP-Net captures features across various scales, enabling effective segmentation of both large and small cracks patterns.
- Refined Edge Detection: The enhanced HED Network enhances boundary detection, crucial for accurately segmenting irregular cracks manifestations, leading to better segmentation quality.
- Enhanced Contextual Understanding: By combining feature extraction with edge detection, the model achieves a comprehensive understanding of the image context, which is vital for effective segmentation.

The proposed Hybrid Model effectively segments road cracks images by leveraging the strengths of both the improved PSP-Net and enhanced HED Network, ultimately enhancing the detection and classification of cracks.

3.4 Feature extraction techniques

Feature extraction is a pivotal phase in the proposed methodology, designed to identify and capture the most pertinent information from cracks images, facilitating accurate cracks detection. The approach integrates various techniques to ensure a comprehensive representation of the data. This method not only enhances the robustness of the model but also leverages both low-level and high-level features, enabling the system to effectively differentiate between without crack and road cracks.

3.4.1. LBP-based texture features extraction

LBP are used to describe the texture of the road crack images. LBP is a powerful method for texture classification that captures the spatial structure of local image texture. Following are the mathematical formulations for the LBP based feature extraction [21]:

LBP Calculation: For each pixel (x, y) in the image I, calculate the LBP value by comparing the pixel with its neighbourhood [21]:

$$LBP(x, y) = \sum_{p=0}^{P-1} s\left(I(x_p, y_p) - I(x, y)\right) \cdot 2^p \tag{9}$$

Where:

(x_p, y_p) are the coordinates of the p^{th} neighboring pixel,

P is the number of neighbors,

$s(z)$ is a sign function:

$$s(x) = \begin{cases} 1 & if\ z \geq 0 \\ 0 & otherwise \end{cases} \quad (10)$$

Uniform Patterns: Detect uniform patterns, which have at most two 0-1 or 1-0 transitions in the binary string. The uniform LBP value LBP_{u2} is defined as:

$$LBP_{u2}(x, y) = \begin{cases} \sum_{p=0}^{P-1} s\begin{pmatrix} I(x_p, y_p) - \\ I(x, y) \end{pmatrix} \cdot 2^p & if\ uniform \\ P + 1 & otherwise \end{cases} \quad (11)$$

Histogram Calculation: Compute the histogram of the LBP values over a region R:

$$H_k = \sum_{(x,y)\in R} \delta(LBP_{u2}(x, y), k) \quad (12)$$

Where $\delta(a, b)$ is the Kronecker delta function, which is 1 if $a = b$ and 0 otherwise, and k is a possible LBP value.

The LBP histogram H represents the texture features of the region.

3.4.2. Deep features extraction

Deep feature extraction is achieved through the hybrid architecture of the improved PSP-Net combined with the enhanced HED Network. This approach harnesses the power of deep learning to derive high-level representations from the cracks images, effectively capturing complex patterns indicative of various road cracks.

- Convolution Operation: The feature extraction process primarily relies on the convolutional layers within the Improved PSP-Net. Each layer applies convolutional filters to the input feature map, allowing for the detection of features across multiple spatial hierarchies. The convolution operation is mathematically expressed as [21]:

$$F_{i,j} = \sum_{m,n} W_{m,n} \cdot I_{i+m,j+n} \quad (13)$$

Where:

- $F_{i,j}$ is the output feature at position (i, j).
- $W_{m,n}$ represents the weights of the convolutional filter.
- I is the input image or feature map.

This operation enables the network to capture intricate patterns and structures present in the cracks images, facilitating effective segmentation of cracks -affected areas.

- Multi-Scale Context Aggregation: The Improved PSP-Net utilizes a Pyramid Pooling Module to gather contextual information from different scales, further enhancing feature extraction. This allows the model to capture both global context and local details, critical for understanding the diverse manifestations of road cracks.
- Integration with HED Network: To refine the boundary detection of cracks -affected regions, the output of the Improved PSP-Net is fed into the Enhanced HED Network. This integration focuses on high-precision edge detection, ensuring that fine details and irregularities in cracks boundaries are accurately captured. The mathematical formulation for this process can be expressed as:

$$E_{refined} = \sigma(W_{HED} * E_{fused}) \quad (14)$$

Where:

- $E_{refined}$ is the refined edge map produced by the HED Network.
- W_{HED} represents the weights in the HED architecture.
- E_{fused} is the fused output from the Improved PSP-Net, combining multi-scale features.

- Final Feature Map Generation: The deep features are synthesized from both the Improved PSP-Net and the Enhanced HED Network to create a comprehensive feature map, F_{final}, which captures both texture and high-level patterns necessary for accurate cracks segmentation and classification:

$$F_{final} = \sigma(W_{final} * E_{refined}) \quad (15)$$

Here:

- W_{final} are the weights of the last convolutional layer.
- σ is a non-linear activation function, such as ReLU, which introduces non-linearity, allowing the model to learn complex mappings.

The deep features extracted from the hybrid model complement the semantic segmented features, providing a rich representation of both texture and higher-order patterns in the images, which is essential for the subsequent classification tasks.

3.4.3. Combined feature set

To establish a robust model for the detection of road cracks, it is essential to integrate the deep features extracted from the Improved PSP-Net combined with the Enhanced HED Network, alongside the LBP-based texture features. This approach effectively leverages the strengths of both feature types, providing a comprehensive representation of the image data.

Let f_{deep} denote the feature vector derived from the deep features extracted via the hybrid model of Improved PSP-Net combined with the Enhanced HED Network.

Let f_{LBP} represent the feature vector obtained from the Local Binary Patterns (LBP) method, which captures crucial texture characteristics of the cracks images.

- Normalization: Each feature vector is normalized to ensure consistent scaling, transforming them to have zero mean and unit variance. This can be expressed mathematically as:

$$f_i^{norm} = \frac{f_i - \mu_i}{\sigma_i} \tag{16}$$

Where:

- f_i is a feature vector (e.g., f_{deep} or f_{LBP})
- μ_i is the mean of f_i
- σ_i is the standard deviation of f_i.

- Concatenation: After normalization, the normalized feature vectors are concatenated to form a single, comprehensive feature vector $f_{Combined}$, which integrates both deep and texture features. This is represented mathematically as:

$$f_{Combined} = \left[f_{deep}^{norm} \parallel f_{LBP}^{norm}\right] \tag{17}$$

Where $\parallel$ denotes the concatenation operation.

The resulting combined feature set enhances the model's capability to accurately detect and classify road cracks by integrating the rich information captured by both the deep features and the LBP-based texture features.

3.5 MRMR based feature selection

To optimize the combined feature set for road crack detection, the MRMR (Minimum Redundancy Maximum Relevance) feature selection method is employed. This approach aims to refine the feature set by selecting the most informative features while minimizing redundancy among them, ultimately enhancing the classification performance [23].

The MRMR feature selection process is applied to the combined feature set $f_{Combined}$ to identify and retain the most relevant features for the cracks classification task.

- Relevance Calculation: The relevance of each feature f_1 in $f_{Combined}$ with respect to the target variable Y is computed using mutual information:

$$R_j = I(f_j; Y) \tag{18}$$

Here, R_j signifies the relevance score of feature f_j.

- Redundancy Calculation: The redundancy between any two features f_i and f_j in $f_{Combined}$ is assessed by their mutual information:

$$D_{i,j} = I(f_i; f_j) \tag{19}$$

The overall redundancy for a selected subset S of features can be quantified as:

$$D(S) = \frac{1}{|S|^2} \sum_{i,j \in S} I(f_i; f_j) \tag{20}$$

- Objective Function: The objective function for MRMR feature selection is defined to maximize the trade-off between relevance and redundancy:

$$J(s) = \frac{1}{|S|} \sum_{f_j \in S} R_j - \lambda D(S) \tag{21}$$

In this equation, λ is a weighting factor that balances the importance of relevance versus redundancy.

- Selected Features: The final selected feature set $F_{Selected}$ can be defined as follows:

$$F_{Selected} = \{f_j \mid j \in S\} \tag{22}$$

Where S is the index set of selected features based on the MRMR criteria.

The application of MRMR-based feature selection on the combined feature set $f_{Combined}$ ensures that the model retains the most significant features for detecting and classifying road cracks. By focusing on relevant and non-redundant features, this refined subset enhances the accuracy and efficiency of the cracks classification model.

3.6 Classification using XGBoost classifier

XGBoost operates by building an ensemble of decision trees, where each tree corrects the residuals of the previous trees [24].

Objective Function: XGBoost minimizes the objective function, which combines a loss function l and a regularization term Ω:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{t=1}^{T} \Omega(f_t) \tag{23}$$

Where:

- $l(y_i, \hat{y}_i)$: Loss function (e.g., log-loss for classification).
- $\Omega(f_t) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$: Regularization for controlling model complexity.

Tree Model: Each decision tree $f_t(x)$ maps an input x to a leaf weight w:

$$f_t(x) = w_{q(x)} \tag{24}$$

Where $q(x)$ maps x to a leaf node.

Additive Training: New trees sequentially reduce the residuals of the previous model:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \tag{25}$$

Second-Order Approximation: XGBoost uses second-order Taylor expansion to approximate the loss:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \begin{bmatrix} g_i f_t(x_i) + \\ \frac{1}{2} h_i f_t^2(x_i) \end{bmatrix} + \Omega(f_t) \tag{26}$$

Where:

- $g_i = \frac{\partial l}{\partial \hat{y}_i^{(t-1)}}$: Gradient.
- $h_i = \frac{\partial^2 l}{\partial \left(\hat{y}_i^{(t-1)}\right)^2}$: Hessian.

Split Gain: The quality of a split is measured by the gain:

$$Gain = \frac{1}{2} \begin{bmatrix} \frac{\left(\sum_{i \in left} g_i\right)^2}{\sum_{i \in left} h_i + \lambda} + \\ \frac{\left(\sum_{i \in right} g_i\right)^2}{\sum_{i \in right} h_i + \lambda} - \\ \frac{\left(\sum_{i \in parent} g_i\right)^2}{\sum_{i \in parent} h_i + \lambda} \end{bmatrix} - \gamma \tag{27}$$

Key Hyperparameters

- Learning Rate (η): Controls the contribution of each tree.
- Max Depth (d): Limits tree depth to balance accuracy and generalization.
- Subsample Ratio: Proportion of samples used for training each tree.
- Regularization (λ, γ): Prevent overfitting.

Output: The classifier predicts the class c_k with the highest probability:

$$\hat{y}_i = \arg\max_k P(y_i = c_k | x_i) \tag{28}$$

This formulation ensures effective classification of road cracks with high accuracy and robustness.

4. Results and discussion

4.1 Datasets

Crack Tree: The Crack Tree dataset focuses on the intricate and often complex patterns formed by cracks on pavement surfaces [25]. This dataset is named for the tree-like branching patterns that cracks can form, which pose unique challenges for detection algorithms. The images within this dataset highlight such patterns, providing a robust testing ground for models that aim to identify and map these branching cracks accurately. The Crack Tree dataset helps in advancing methods that can handle more complex crack geometries effectively.

Deep Crack: Deep Crack is a dataset tailored for deep learning applications in crack detection [25]. It contains a large number of images with varied crack patterns, annotated with precision to facilitate the training of deep neural networks. The dataset supports the development of deep learning models by providing diverse and richly annotated data, ensuring that models trained on this dataset can generalize well to real-world scenarios. The emphasis on deep learning makes this dataset pivotal for modern approaches in automated crack detection.

Crack 500: Crack 500 is another crucial dataset in the realm of pavement crack detection research [25]. It includes a selection of 500 images, each capturing different types of cracks on various pavement surfaces. The images in this dataset have been manually labeled to identify the cracks, ensuring high accuracy in the ground truth data. The dataset's moderate size and well-annotated images provide a balanced platform for both training machine learning models and evaluating their performance in detecting pavement cracks.

4.2 Evaluation parameters

Here are the evaluation metrics employed in the analysis

Table 1. Illustrations of sample images taken from the dataset [25]

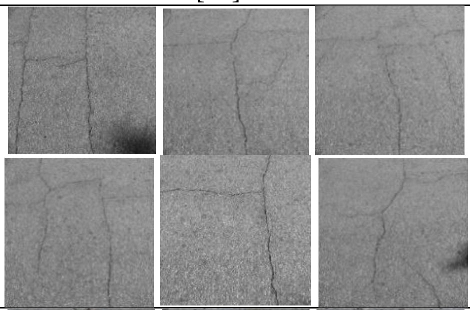
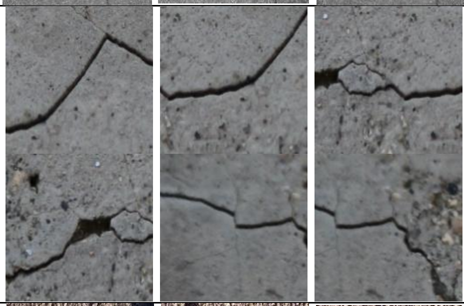
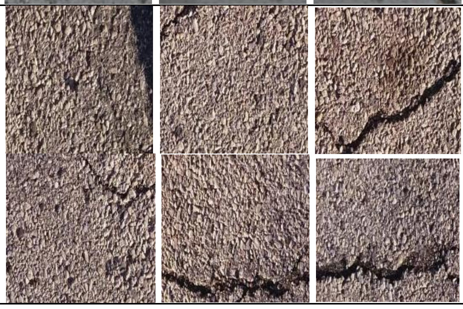
Crack tree	
Deep crack	
Crack 500	

Table 2: Segmented output for the input road crack image

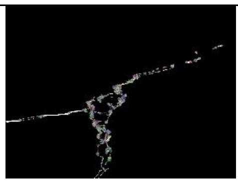
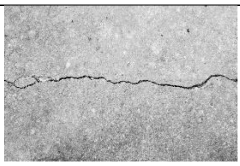
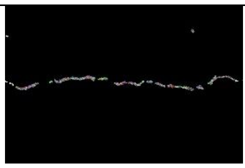
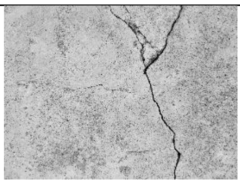
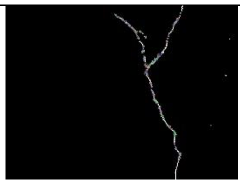
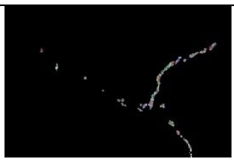
Input Image	Segmented Image
	
	
	
	

Table 3. Image enhancement and segmentation parameters

Parameters Image Enhancement	**Avg value Crack 500**	**Avg value Crack tree**	**Avg value Deep Crack**
EC (Edge Content)	2.56	3.45	2.58
EME (Enhancement Measure of Emphasis)	1.045	2.03	1.034
Image Segmentation Parameters			
NCC (Normalized Cross Correlation)	0.907	0.816	0.9234
MI (Mutual Information)	5.032	4.394	4.848
BDE (Boundary Displacement Error)	4.980	4.30	4.10
FSIM (Feature Similarity Index)	0.8729	0.815	0.869

Figure. 3 Confusion matrix plot of proposed road crack detection for crack 500 dataset

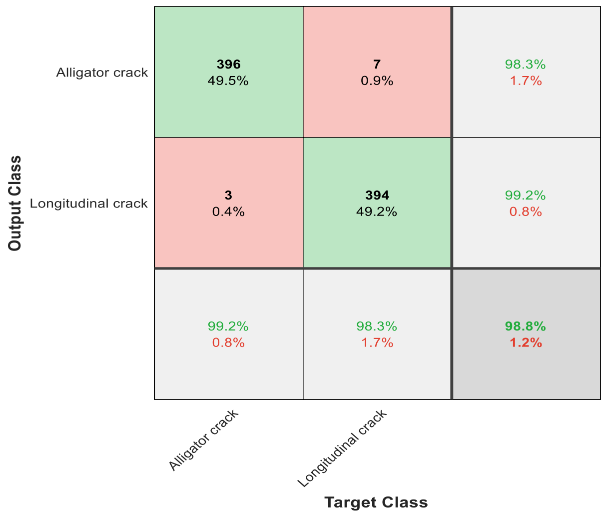


Figure. 4 Confusion matrix plot of proposed road crack detection for crack tree dataset

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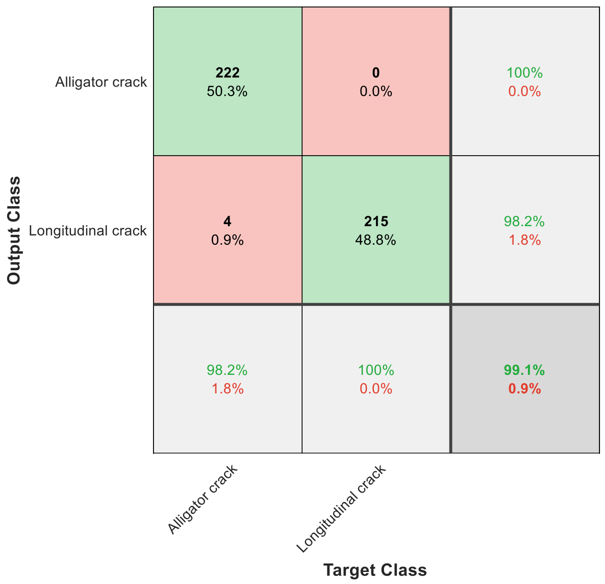

Figure. 5 Confusion matrix plot of proposed road crack detection for crack forest dataset

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (29)$$

$$Precision = \frac{TP}{TP+FP} \quad (30)$$

$$Sensitivity = \frac{TP}{TP+FN} \quad (31)$$

$$Specificity = \frac{TN}{TN+FN} \quad (32)$$

$$ErrorRate = \frac{FP+FN}{TP+TN+FP+FN} \quad (33)$$

$$False\ Positive\ Rate(FPR) = \frac{FP}{FP+TN} \quad (34)$$

$$F-Score = \frac{2TP}{2TP+FP+FN} \quad (35)$$

$$Matthews\ Correlation\ Coefficient\ (MCC) = \frac{(TP\times TN)-(FP\times FN)}{\sqrt{(TP+FN)(TP+FP)(TN+FN)(TN+FP)}} \quad (36)$$

$$Kappa\ Statistics = \frac{2(TP\times TN-FN\times FP)}{(TP+FP)\times(FP+TN)+(TN+FN)\times(FN+TN)} \quad (37)$$

4.3 Results

The analysis of the results indicates distinct performance variations across the three methods: Crack 500, Crack Tree, and Deep Crack, for both image enhancement and segmentation tasks. In terms of edge content (EC), Crack Tree demonstrates the highest value (3.45), signifying better edge definition, followed by Deep Crack (2.58) and Crack 500 (2.56). Similarly, for the Enhancement Measure of Emphasis (EME), Crack Tree again outperforms with 2.03, while Crack 500 (1.045) and Deep Crack (1.034) show comparable but lower values, highlighting Crack Tree's superior image enhancement capabilities.

For image segmentation, Deep Crack achieves the best performance in normalized cross-correlation (NCC) with 0.9234, reflecting high alignment accuracy. Crack 500 follows with 0.907, while Crack Tree scores the lowest at 0.816. In terms of mutual information (MI), Crack 500 leads with 5.032, followed by Deep Crack (4.848) and Crack Tree (4.394), indicating better feature correlation for Crack 500. For boundary displacement error (BDE), Deep Crack records the lowest value (4.10), suggesting precise boundary segmentation, outperforming Crack Tree (4.30) and Crack 500 (4.980). Lastly, for feature similarity index (FSIM), Crack 500 performs best (0.8729), slightly ahead of Deep Crack (0.869), while Crack Tree lags behind (0.815). Overall, each method exhibits unique strengths, with Crack Tree excelling in enhancement metrics and Deep Crack showing robust segmentation accuracy.

Table 4. Comparative Performance Analysis of Road Crack Detection Model across Different Datasets

Metric	Crack 500 Dataset	Crack Tree Dataset	Crack forest Dataset
Accuracy	97.73%	98.75%	99.09%
Error	2.27%	1.25%	0.91%
Sensitivity	99.53%	99.25%	98.23%
Specificity	96.04%	98.25%	100.00%
Precision	95.95%	98.26%	100.00%
False Positive Rate	3.96%	1.75%	0.00%
F1 Score	97.71%	98.75%	99.11%
Matthews Correlation Coefficient	95.53%	97.50%	98.20%
Kappa	95.47%	97.50%	98.19%

Table 4 showcases the model's performance across three datasets, with accuracy improving from 97.73% (Crack 500) to 99.09% (Crack Forest). Key metrics like specificity and precision reach 100% for Crack Forest, highlighting excellent reliability and minimal false positives.

Table 5 presents the Fold-5 cross-validation performance of the proposed recognition algorithm across different segmentation methods for three datasets. The results show a consistent improvement

Table 5. Tabular Results for Fold-5 Cross Validation Performance of Proposed Recognition Algorithm for Various Datasets

Metric	Crack 500 Dataset			Crack Tree Dataset			Crack forest Dataset		
	IPSP-Net Segmentation	**EHED-NET IPSP-Net Segmentation**	**Hybrid Segmentation + MRMR**	**IPSP-Net Segmentation**	**EHED-NET IPSP-Net Segmentation**	**Hybrid Segmentation + MRMR**	**IPSP-Net Segmentation**	**EHED-NET IPSP-Net Segmentation**	**Hybrid Segmentation + MRMR**
Accuracy	96.50%	97.50%	98.00%	95.00%	96.00%	96.50%	92.50%	94.40%	94.90%
Error	3.50%	2.50%	2.00%	5.00%	4.00%	3.50%	7.50%	5.60%	5.10%
Sensitivity	94.50%	96.00%	97.50%	93.50%	94.50%	95.00%	89.00%	91.50%	92.50%
Specificity	98.00%	98.50%	99.00%	96.50%	97.00%	98.00%	96.00%	96.50%	97.50%
Precision	95.00%	97.00%	98.00%	93.00%	94.00%	95.50%	91.00%	89.50%	92.00%
False Positive Rate	2.00%	1.50%	1.00%	3.50%	3.00%	2.00%	4.00%	3.50%	2.50%
F1 Score	94.75%	96.50%	97.75%	93.25%	94.25%	95.25%	90.00%	90.00%	92.00%
MCC	91.50%	93.00%	94.50%	90.00%	92.00%	93.00%	86.00%	89.00%	91.20%
Kappa	84.50%	87.00%	89.00%	82.00%	84.00%	86.00%	79.00%	81.00%	84.00%

Table 6. Comparative analysis of proposed work with previous research works

Ref. No.	**Dataset**	**Method**	**Accuracy**	**Recall / Sensitivity**	**Precision**	**F1-Score**
[11]	Moroccan road image dataset	CNN	--	93.19%	93.19%	93.19%
		VGG-19	--	90.76%	90.76%	90.76%
[12]	Crack 500 Dataset	CNN	71.00%	--	--	--
[13]	Crack Tree Dataset	YOLO V2	--	76.38%	78.65%	77.49%
[14]	Crack Forest Dataset	Optimized YOLOv8	--	59.9%	68.8%	--
[15]	Crack Forest Dataset	Deep convolutional neural network	--	--	--	87%
[16]	Crack 500 Dataset	Basnet + U-multiscale dilated network	--	70%	71.32%	70.65%
[17]	Crack Forest Dataset	Conditional Wasserstein Generative Adversarial network and connectivity maps	--	87.75%	96.79%	91.96%
[18]	AigleRN database	U-Hierarchical Dilated Network	92.7%	93.1%	92.1%	92.4%
Proposed Work	**Crack 500 Dataset**	**Hybrid Segmentation + MRMR**	**97.73%**	99.53%	95.95%	97.71%
Proposed Work	**Crack Tree Dataset**	**Hybrid Segmentation + MRMR**	**98.75%**	99.25%	98.26%	98.75%
Proposed Work	**Crack Forest Dataset**	**Hybrid Segmentation + MRMR**	**99.09%**	100%	100%	99.11%

in performance as we move from IPSP-Net to Hybrid Segmentation + MRMR across all datasets. For instance, the accuracy ranged from 92.50% (Crack Forest with IPSP-Net) to 98.00% (Crack 500 with Hybrid Segmentation + MRMR), with similar trends observed in sensitivity, specificity, and F1 scores. The false positive rates were lowest for Hybrid Segmentation + MRMR, and the Matthews

Correlation Coefficient (MCC) and Kappa also demonstrated higher values for this method, indicating a robust recognition model across different dataset types.

Table 6 compares the proposed crack detection method with several other existing methods on three datasets (Crack 500, Crack Tree, and Crack Forest). Methods are compared in terms of their accuracy, recall/sensitivity, precision, and F1-score in the table above. For example, [11] achieves 89.9% accuracy using CNN; [12] has 93.19% accuracy using VGG-19; while [13] combined multiple object detection methods (YOLO V2, optimized YOLOv8) on the Kaggle and MIAS datasets, achieving precision 0.9640, recall 0.92. By contrast, the proposed approach outperforms all features domain in the mentioned datasets and the accurate results for Crack 500, Crack Tree and Crack Forest datasets are: 97.73%, 98.75% and 99.09%. It also outperforms in recall/sensitivity (99.53%, 99.25%, 100%) and precision (95.95%, 98.26%, and 100%), giving an F1-scores of 97.71%, 98.75%, and 99.11%, respectively. Such comparison highlights the unambiguity and great performance of the proposed hybrid segmentation methodology with MRMR feature selection, which itself, is an improvement over the earlier techniques.

5. Conclusion

The proposed road crack detection model presented in this paper demonstrates remarkable advancements in both accuracy and robustness when compared to existing methods. By combining an Improved PSP-Net, Enhanced HED Network, and LBP-based texture features, the model effectively captures multiscale contextual information and fine-grained surface details, enabling precise crack localization under diverse conditions. The model's performance on three distinct datasets (Crack 500, Crack Tree, and Crack Forest) further underscores its capability, with accuracy rates reaching as high as 99.09%. This exceeds the performance of prior state-of-the-art methods, highlighting the proposed model's superior ability to detect complex crack patterns. Moreover, the results from cross-validation and segmentation evaluation indicate a consistent improvement in detection reliability, with the hybrid segmentation + MRMR approach yielding the best results in terms of sensitivity, specificity, and F1 scores. Overall, the proposed method proves to be a highly effective solution for road crack detection, making it suitable for real-world road maintenance and inspection applications. Future work could focus on further improving model efficiency for real-time applications and expanding its generalization across more varied environmental conditions and road surfaces.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

Dr. Shubhlakshmi Tiwari, as the principal author, led the study design, data acquisition, analysis, and manuscript preparation. V Somesh, Mrs. Rolly V Joshi, and Mr. Gyan Prakash Sharma assisted with the experimental procedures, data analysis, and manuscript revision, providing valuable feedback to improve the research.

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